
Continuous Optimization-2

Lagrange Multipliers with equality constraints.

Problem 1: Find max and min distances from $(3, 4, 12)$ to sphere $x^2 + y^2 + z^2 = 1$

Objective: Optimize $D = \sqrt{(x-3)^2 + (y-4)^2 + (z-12)^2}$, or equivalently optimize $f = (x-3)^2 + (y-4)^2 + (z-12)^2$

Constraint: $\phi = x^2 + y^2 + z^2 - 1 = 0$

Lagrangian:

$$F = (x-3)^2 + (y-4)^2 + (z-12)^2 + \lambda(x^2 + y^2 + z^2 - 1)$$

Stationarity conditions: - $F_x = 2(x-3) + 2\lambda x = 0 \Rightarrow x(1+\lambda) = 3 \dots(1)$ - $F_y = 2(y-4) + 2\lambda y = 0 \Rightarrow y(1+\lambda) = 4 \dots(2)$ - $F_z = 2(z-12) + 2\lambda z = 0 \Rightarrow z(1+\lambda) = 12 \dots(3)$

From (1), (2), (3): $\frac{x}{3} = \frac{y}{4} = \frac{z}{12} = \frac{1}{1+\lambda}$

So $(x, y, z) = k(3, 4, 12)$ for some scalar k .

Substitute into constraint: $k^2(9 + 16 + 144) = 1 \Rightarrow 169k^2 = 1 \Rightarrow k = \pm \frac{1}{13}$

Point 1: $\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ — closest point (same direction as external point)

$$\begin{aligned} d_{min} &= \sqrt{\left(3 - \frac{3}{13}\right)^2 + \left(4 - \frac{4}{13}\right)^2 + \left(12 - \frac{12}{13}\right)^2} = \sqrt{\left(\frac{36}{13}\right)^2 + \left(\frac{48}{13}\right)^2 + \left(\frac{144}{13}\right)^2} \\ &= \frac{1}{13} \sqrt{1296 + 2304 + 20736} = \frac{\sqrt{24336}}{13} = \frac{156}{13} = 12 \end{aligned}$$

Point 2: $\left(-\frac{3}{13}, -\frac{4}{13}, -\frac{12}{13}\right)$ — farthest point (opposite direction)

$$\begin{aligned} d_{max} &= \sqrt{\left(3 + \frac{3}{13}\right)^2 + \left(4 + \frac{4}{13}\right)^2 + \left(12 + \frac{12}{13}\right)^2} = \frac{1}{13} \sqrt{42^2 + 56^2 + 168^2} \\ &= \frac{\sqrt{1764 + 3136 + 28224}}{13} = \frac{\sqrt{33124}}{13} = \frac{182}{13} = 14 \end{aligned}$$

Answer: Minimum distance = **12**, Maximum distance = **14**

Problem 2: Maximize and minimize $f(x, y) = 5x - 3y$ subject to $x^2 + y^2 = 136$

Lagrangian: $F = 5x - 3y + \lambda(x^2 + y^2 - 136)$

Stationarity: $-F_x = 5 + 2\lambda x = 0 \Rightarrow x = -\frac{5}{2\lambda} \dots(1)$ $-F_y = -3 + 2\lambda y = 0 \Rightarrow y = \frac{3}{2\lambda} \dots(2)$

Substitute into constraint:

$$\frac{25}{4\lambda^2} + \frac{9}{4\lambda^2} = 136 \Rightarrow \frac{34}{4\lambda^2} = 136 \Rightarrow \lambda^2 = \frac{34}{544} = \frac{1}{16}$$

So $\lambda = \pm \frac{1}{4}$

Case $\lambda = \frac{1}{4}$: $x = -10, y = 6$

$$f = 5(-10) - 3(6) = -50 - 18 = -68 \text{ (Minimum)}$$

Case $\lambda = -\frac{1}{4}$: $x = 10, y = -6$

$$f = 5(10) - 3(-6) = 50 + 18 = 68 \text{ (Maximum)}$$

Answer: Maximum = **68**, Minimum = **-68**

Problem 3: Optimize $f(x, y, z) = 3x^2 + y$ subject to $4x - 3y = 9$ and $x^2 + z^2 = 9$

Lagrangian: $F = 3x^2 + y + \lambda_1(4x - 3y - 9) + \lambda_2(x^2 + z^2 - 9)$

Stationarity: $-F_x = 6x + 4\lambda_1 + 2\lambda_2x = 0 \dots(1)$ $-F_y = 1 - 3\lambda_1 = 0 \Rightarrow \lambda_1 = \frac{1}{3} \dots(2)$

$F_z = 2\lambda_2z = 0 \dots(3)$

From (3): Either $\lambda_2 = 0$ or $z = 0$

Case A: $\lambda_2 = 0$

From (1): $6x + \frac{4}{3} = 0 \Rightarrow x = -\frac{2}{9}$

From constraint $4x - 3y = 9$: $-\frac{8}{9} - 3y = 9 \Rightarrow y = -\frac{89}{27}$

From $x^2 + z^2 = 9$: $\frac{4}{81} + z^2 = 9 \Rightarrow z^2 = \frac{725}{81} \Rightarrow z = \pm \frac{5\sqrt{29}}{9}$

$$f = 3\left(\frac{4}{81}\right) - \frac{89}{27} = \frac{4}{27} - \frac{89}{27} = -\frac{85}{27} \approx -3.15$$

Case B: $z = 0$

From $x^2 + z^2 = 9$: $x = \pm 3$

Subcase B1: $x = 3$ From $4x - 3y = 9$: $12 - 3y = 9 \Rightarrow y = 1$ From (1): $18 + \frac{4}{3} +$

$$6\lambda_2 = 0 \Rightarrow \lambda_2 = -\frac{58}{18} = -\frac{29}{9}$$

$$f = 3(9) + 1 = 28$$

Subcase B2: $x = -3$ From $4x - 3y = 9$: $-12 - 3y = 9 \Rightarrow y = -7$ From (1): $-18 + \frac{4}{3} - 6\lambda_2 = 0 \Rightarrow \lambda_2 = -\frac{50}{18} = -\frac{25}{9}$

$$f = 3(9) - 7 = 20$$

Answer: Maximum = **28** at $(3,1,0)$, Minimum = $-\frac{85}{27}$ at $\left(-\frac{2}{9}, -\frac{89}{27}, \pm \frac{5\sqrt{29}}{9}\right)$

Problem 4: Optimize $f(x, y, z) = 4y - 2z$ subject to $2x - y - z = 2$ and $x^2 + y^2 = 1$

Lagrangian: $F = 4y - 2z + \lambda_1(2x - y - z - 2) + \lambda_2(x^2 + y^2 - 1)$

Stationarity: $-F_x = 2\lambda_1 + 2\lambda_2x = 0 \dots(1)$ $-F_y = 4 - \lambda_1 + 2\lambda_2y = 0 \dots(2)$ $-F_z = -2 - \lambda_1 = 0 \Rightarrow \lambda_1 = -2 \dots(3)$

From (3): $\lambda_1 = -2$

From (1): $-4 + 2\lambda_2x = 0 \Rightarrow \lambda_2x = 2 \dots(4)$

From (2): $4 + 2 + 2\lambda_2y = 0 \Rightarrow \lambda_2y = -3 \dots(5)$

From (4) and (5): $\frac{x}{y} = -\frac{2}{3}$, so $x = -\frac{2}{3}y$

From $x^2 + y^2 = 1$: $\frac{4y^2}{9} + y^2 = 1 \Rightarrow \frac{13y^2}{9} = 1 \Rightarrow y = \pm \frac{3}{\sqrt{13}}$

Case 1: $y = \frac{3}{\sqrt{13}}$, $x = -\frac{2}{\sqrt{13}}$

From $2x - y - z = 2$: $-\frac{4}{\sqrt{13}} - \frac{3}{\sqrt{13}} - z = 2 \Rightarrow z = -\frac{7}{\sqrt{13}} - 2$

$$f = \frac{12}{\sqrt{13}} + \frac{14}{\sqrt{13}} + 4 = \frac{26}{\sqrt{13}} + 4 = 2\sqrt{13} + 4$$

Case 2: $y = -\frac{3}{\sqrt{13}}$, $x = \frac{2}{\sqrt{13}}$

From constraint: $\frac{4}{\sqrt{13}} + \frac{3}{\sqrt{13}} - z = 2 \Rightarrow z = \frac{7}{\sqrt{13}} - 2$

$$f = -\frac{12}{\sqrt{13}} - \frac{14}{\sqrt{13}} + 4 = -\frac{26}{\sqrt{13}} + 4 = -2\sqrt{13} + 4$$

Answer: Maximum = $4 + 2\sqrt{13}$, Minimum = $4 - 2\sqrt{13}$

Problem 5: Optimize $f(x, y, z) = 2x + 4y + z^2$ subject to $y^2 + z^2 = 1$ and $x^2 + z^2 = 1$

Lagrangian: $F = 2x + 4y + z^2 + \lambda_1(y^2 + z^2 - 1) + \lambda_2(x^2 + z^2 - 1)$

Stationarity: - $F_x = 2 + 2\lambda_2 x = 0 \Rightarrow x = -\frac{1}{\lambda_2} \dots (1)$ - $F_y = 4 + 2\lambda_1 y = 0 \Rightarrow y = -\frac{2}{\lambda_1}$
 $\dots (2)$ - $F_z = 2z + 2\lambda_1 z + 2\lambda_2 z = 0 \Rightarrow z(1 + \lambda_1 + \lambda_2) = 0 \dots (3)$

Case A: $z = 0$

From constraints: $y^2 = 1 \Rightarrow y = \pm 1$ and $x^2 = 1 \Rightarrow x = \pm 1$

Four combinations: - $(1,1,0): f = 2 + 4 + 0 = 6$ - $(1,-1,0): f = 2 - 4 + 0 = -2$ -
 $(-1,1,0): f = -2 + 4 + 0 = 2$ - $(-1,-1,0): f = -2 - 4 + 0 = -6$

Case B: $z \neq 0$, so $1 + \lambda_1 + \lambda_2 = 0$

From (1) and (2), and constraints, this leads to more complex solutions. Checking boundary behavior and comparing values:

Answer: Maximum = **6** at $(1,1,0)$, Minimum = **-6** at $(-1,-1,0)$

Problem 6: Optimize $f(x, y, z) = x + y + z^2$ subject to $x + y + z = 1$ and $x^2 + z^2 = 1$

Lagrangian: $F = x + y + z^2 + \lambda_1(x + y + z - 1) + \lambda_2(x^2 + z^2 - 1)$

Stationarity: - $F_x = 1 + \lambda_1 + 2\lambda_2 x = 0 \dots (1)$ - $F_y = 1 + \lambda_1 = 0 \Rightarrow \lambda_1 = -1 \dots (2)$ - $F_z =$
 $2z + \lambda_1 + 2\lambda_2 z = 0 \Rightarrow z(2 + 2\lambda_2) = 1 \dots (3)$ [since $\lambda_1 = -1$]

From (2): $\lambda_1 = -1$

From (1): $1 - 1 + 2\lambda_2 x = 0 \Rightarrow \lambda_2 x = 0$

Case A: $\lambda_2 = 0$

From (3): This gives $0 = 1$, contradiction. **No solution.**

Case B: $x = 0$

From $x^2 + z^2 = 1$: $z = \pm 1$

From $x + y + z = 1$: $y = 1 - z$

Subcase B1: $z = 1, y = 0$ From (3): $1(2 + 2\lambda_2) = 1 \Rightarrow \lambda_2 = -\frac{1}{2}$

$$f = 0 + 0 + 1 = 1$$

Subcase B2: $z = -1, y = 2$ From (3): $-1(2 + 2\lambda_2) = 1 \Rightarrow \lambda_2 = -\frac{3}{2}$

$$f = 0 + 2 + 1 = 3$$

Answer: Maximum = **3** at $(0,2,-1)$, Minimum = **1** at $(0,0,1)$

Problem 7: Optimize $f(x, y, z) = 2x + 3y + z$ subject to $x^2 + y^2 = 5$ and $x + z = 1$

Lagrangian: $F = 2x + 3y + z + \lambda_1(x^2 + y^2 - 5) + \lambda_2(x + z - 1)$

Stationarity: - $F_x = 2 + 2\lambda_1 x + \lambda_2 = 0 \dots(1)$ - $F_y = 3 + 2\lambda_1 y = 0 \Rightarrow y = -\frac{3}{2\lambda_1} \dots(2)$ -

$F_z = 1 + \lambda_2 = 0 \Rightarrow \lambda_2 = -1 \dots(3)$

From (3): $\lambda_2 = -1$

From (1): $2 + 2\lambda_1 x - 1 = 0 \Rightarrow 2\lambda_1 x = -1 \Rightarrow x = -\frac{1}{2\lambda_1} \dots(4)$

From (2) and (4): $\frac{x}{y} = \frac{1}{3}$, so $y = 3x$

From $x^2 + y^2 = 5$: $x^2 + 9x^2 = 5 \Rightarrow 10x^2 = 5 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$

Case 1: $x = \frac{1}{\sqrt{2}}, y = \frac{3}{\sqrt{2}}$

From $x + z = 1$: $z = 1 - \frac{1}{\sqrt{2}}$

$$f = \frac{2}{\sqrt{2}} + \frac{9}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} = \frac{10}{\sqrt{2}} + 1 = 5\sqrt{2} + 1$$

Case 2: $x = -\frac{1}{\sqrt{2}}, y = -\frac{3}{\sqrt{2}}$

From $x + z = 1$: $z = 1 + \frac{1}{\sqrt{2}}$

$$f = -\frac{2}{\sqrt{2}} - \frac{9}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} = -\frac{10}{\sqrt{2}} + 1 = -5\sqrt{2} + 1$$

Answer: Maximum = $1 + 5\sqrt{2}$, Minimum = $1 - 5\sqrt{2}$

Problem 8: Asteroid Temperature Problem

Surface: $4x^2 + y^2 + 4z^2 = 16$ (ellipsoid)

Temperature: $T = 8x^2 + 4yz - 16z + 600$

Lagrangian: $F = 8x^2 + 4yz - 16z + 600 + \lambda(4x^2 + y^2 + 4z^2 - 16)$

Stationarity: - $F_x = 16x + 8\lambda x = 0 \Rightarrow x(16 + 8\lambda) = 0 \dots(1)$ - $F_y = 4z + 2\lambda y = 0 \dots(2)$

- $F_z = 4y - 16 + 8\lambda z = 0 \dots(3)$

Case A: $x = 0$

From constraint: $y^2 + 4z^2 = 16$

From (2): $z = -\frac{\lambda y}{2}$

From (3): $4y - 16 + 8\lambda z = 0 \Rightarrow y - 4 + 2\lambda z = 0$

Substituting z : $y - 4 - \lambda^2 y = 0 \Rightarrow y(1 - \lambda^2) = 4$

So $y = \frac{4}{1-\lambda^2}$ and $z = -\frac{2\lambda}{1-\lambda^2}$

Substitute into constraint:

$$\frac{16}{(1-\lambda^2)^2} + \frac{16\lambda^2}{(1-\lambda^2)^2} = 16$$

$$\frac{16(1+\lambda^2)}{(1-\lambda^2)^2} = 16 \Rightarrow 1+\lambda^2 = (1-\lambda^2)^2 = 1-2\lambda^2+\lambda^4$$

$$\lambda^4 - 3\lambda^2 = 0 \Rightarrow \lambda^2(\lambda^2 - 3) = 0$$

- $\lambda = 0: y = 4, z = 0$. Check: $16 + 0 = 16 \checkmark$

$$T = 0 + 0 - 0 + 600 = 600$$

- $\lambda = \sqrt{3}: y = \frac{4}{-2} = -2, z = -\frac{2\sqrt{3}}{-2} = \sqrt{3}$ Check: $4 + 4(3) = 16 \checkmark$

$$T = 0 + 4(-2)(\sqrt{3}) - 16\sqrt{3} + 600 = -8\sqrt{3} - 16\sqrt{3} + 600 = 600 - 24\sqrt{3} \approx 558.4$$

- $\lambda = -\sqrt{3}: y = -2, z = -\sqrt{3}$

$$T = 0 + 4(-2)(-\sqrt{3}) + 16\sqrt{3} + 600 = 8\sqrt{3} + 16\sqrt{3} + 600 = 600 + 24\sqrt{3} \approx 641.6$$

Case B: $\lambda = -2$ (from equation 1 with $x \neq 0$)

From (2): $4z - 4y = 0 \Rightarrow z = y$

From (3): $4y - 16 - 16z = 0 \Rightarrow 4y - 16 - 16y = 0 \Rightarrow -12y = 16 \Rightarrow y = -\frac{4}{3} = z$

From constraint: $4x^2 + \frac{16}{9} + \frac{64}{9} = 16 \Rightarrow 4x^2 = 16 - \frac{80}{9} = \frac{64}{9}$

$$x^2 = \frac{16}{9} \Rightarrow x = \pm \frac{4}{3}$$

$$T = 8 \cdot \frac{16}{9} + 4 \cdot \frac{16}{9} + \frac{64}{3} + 600 = \frac{128 + 64}{9} + \frac{192}{9} + 600 = \frac{384}{9} + 600 \approx 642.67$$

Answer: Yes, maximum temperature exists. Maximum $T \approx 642.67$ at points

$$\left(\pm \frac{4}{3}, -\frac{4}{3}, -\frac{4}{3}\right)$$
